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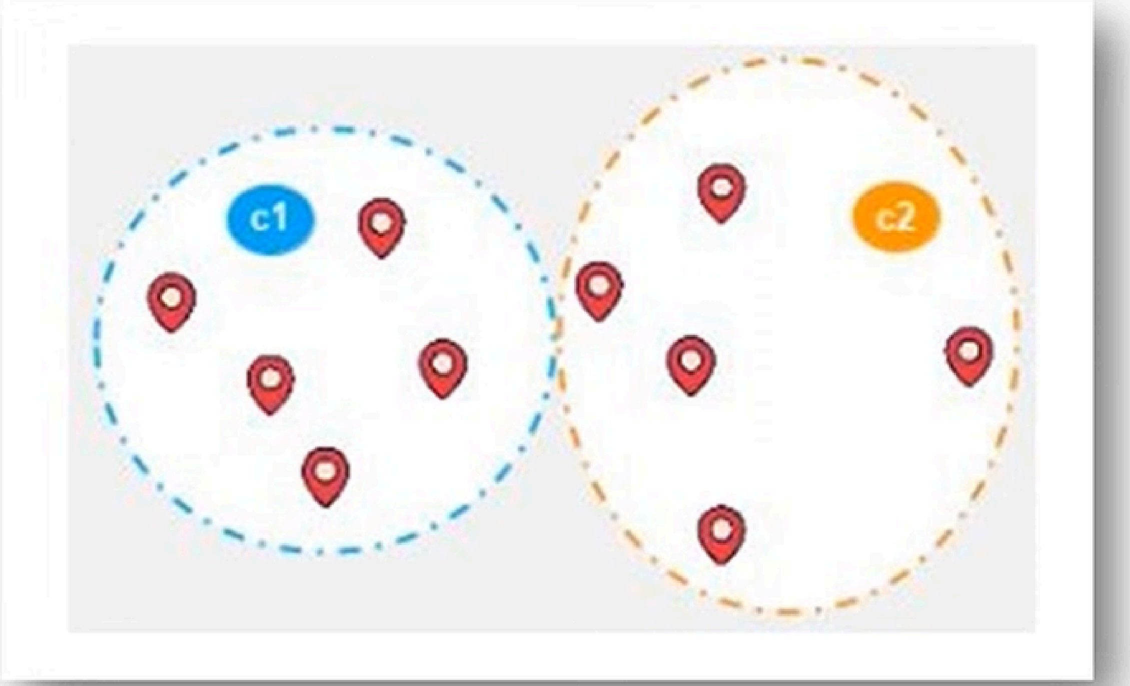


About Clustering

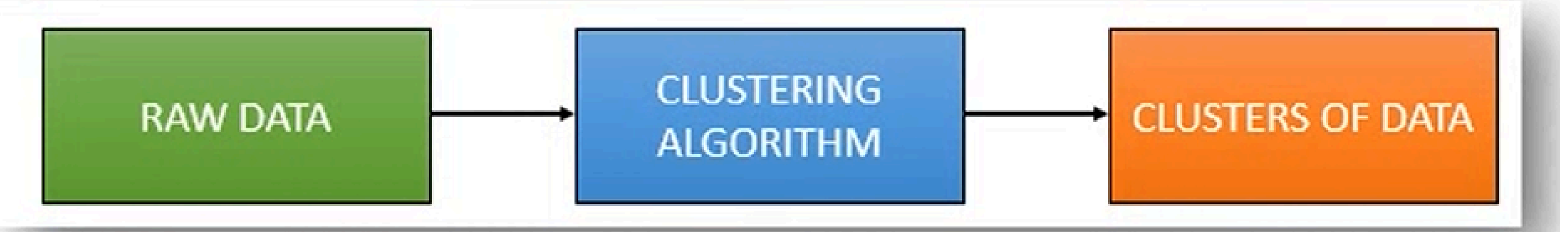
- Clustering is dividing data points into homogeneous classes or clusters:

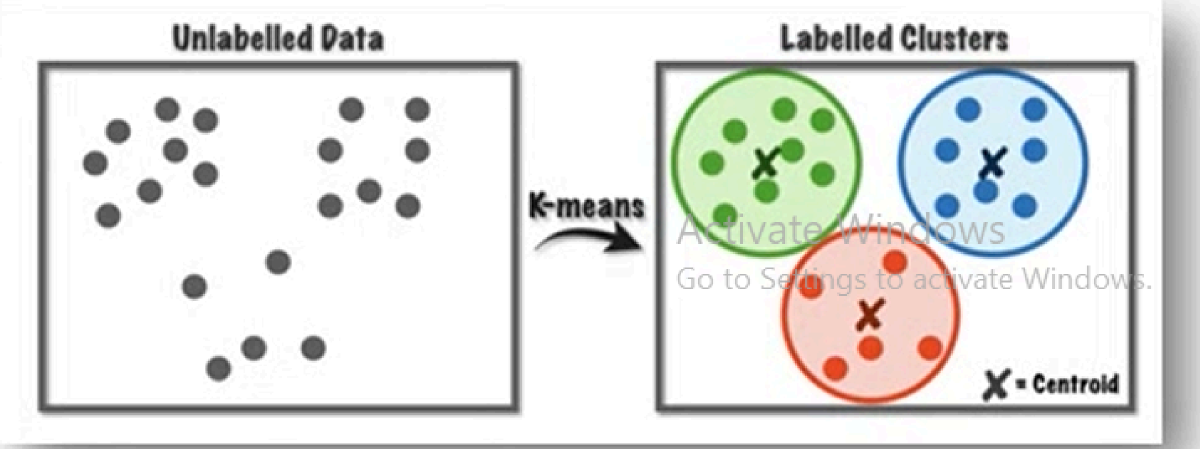
✓Points in the same group are as similar as possible.

✓Points in different group are as dissimilar as possible.

Application of Clustering: Clustering is used in almost all the fields.

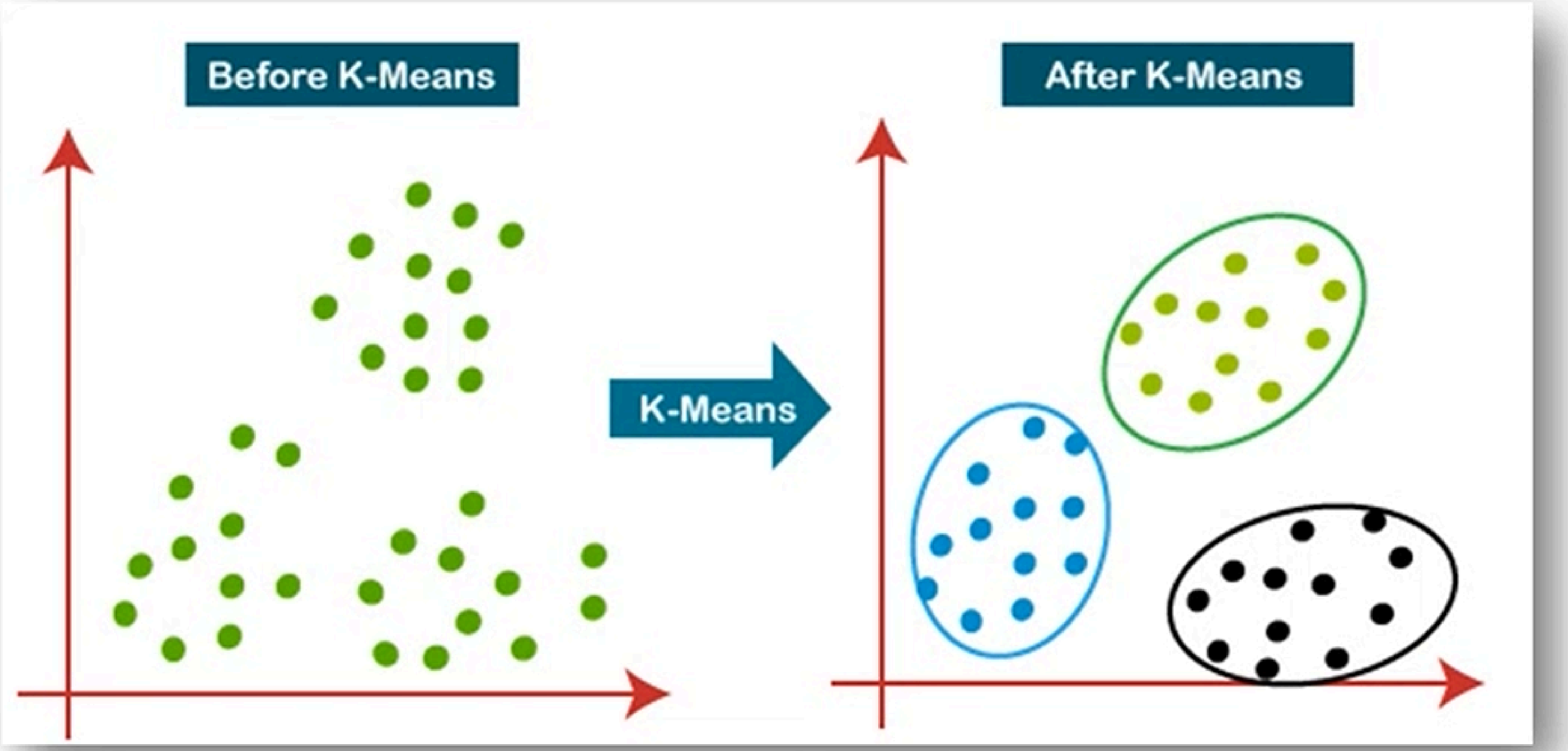
1. Clustering helps in identification of groups of houses on the basis of their value, type and geographical locations.
2. Clustering is used to study earth-quake. Based on the areas hit by an earthquake in a region, clustering can help analyse the next probable location where earthquake can occur.





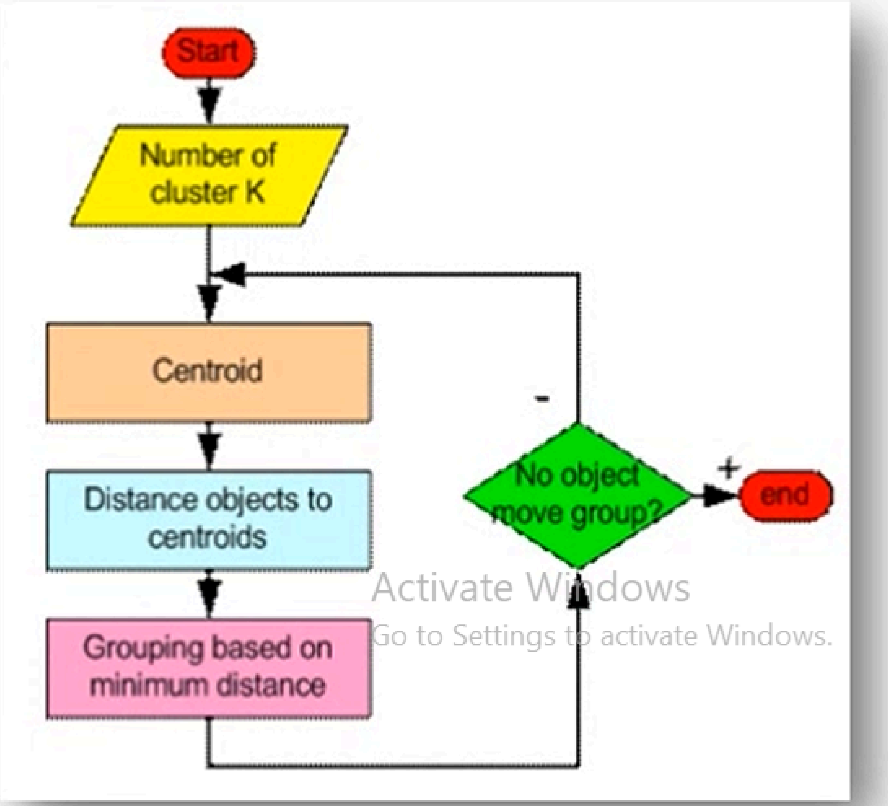
About K-means Clustering

- A K-means clustering algorithm tries to group similar items in the form of clusters.
- The number of groups is represented by K. If K=2, there will be two clusters.
- It is a centroid-based algorithm, where each cluster is associated with a centroid.
- The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.



K-Means Algorithm Steps

- **Step-1:** Select the number K to decide the number of clusters.
- **Step-2:** Select random K points or centroids. (It can be other from the input dataset).
- **Step-3:** Assign each data point to their closest centroid, which will form the predefined K clusters.
- **Step-4:** Calculate the variance and place a new centroid of each cluster.
- **Step-5:** Repeat the third steps, which means reassign each data point to the new closest centroid of each cluster.
- **Step-6:** If any reassignment occurs, then go to step-4 else go to FINISH.
- **Step-7**: The model is ready.

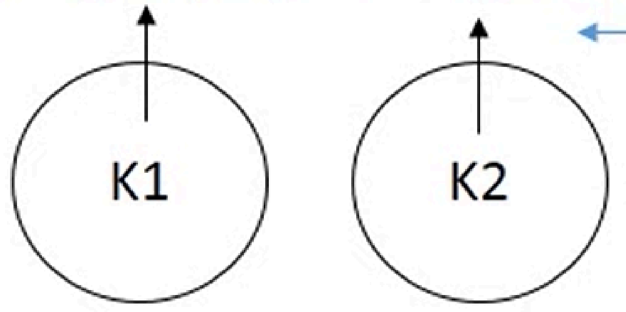


Example

Sr. No.	Height	Weight	Clusters
1	185	72	K1
2	170	56	K2
3	168	60	K2
4	179	68	K1
5	182	72	?
6	188	77	?
7	180	71	?
8	180	70	?
9	183	84	?
10	180	88	?
11	180	67	?
12	177	76	?

STEP 1: Decide Centroids

K1: (185,72) **K2:(170,56)**



STEP 2: Find shortest distance

$$K1 = \sqrt{(168-185)^2 + (60-72)^2}$$

K1 = 20.80

$$K2 = \sqrt{(168-170)^2 + (60-56)^2}$$

K2 = 4.48 (Minimum Dist)

STEP 3: New Centroid Calculation

K2: (170+168/2, 60+56/2)

K2: (169,58) (New Centroid)

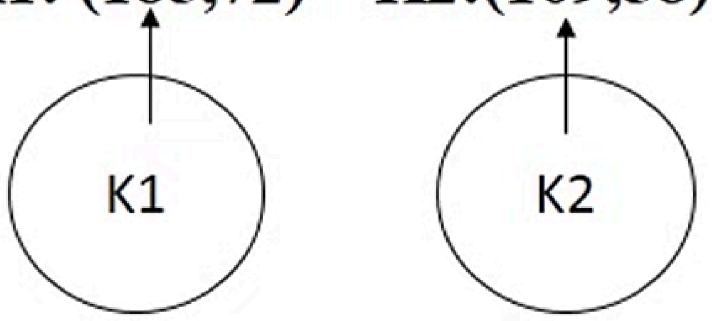
✓ **Euclidean Distance Formula:**

$$d = \sqrt{[(x_o - x_c)^2 + (y_o - y_c)^2]}$$

- X_o, y_o = Observe Valúes
- **Xc. yc** = Centroid Valúes
- **d** = distance

Updated Centroids

K1: (185,72) **K2:(169,58)**



STEP 4: Find shortest distance

$$K1 = \sqrt{(179-185)^2 + (68-72)^2}$$

K1 = 6.32 (Minimum Dist)

$$K2 = \sqrt{(179-169)^2 + (68-58)^2}$$

K2 = 14.14

Remaining Steps:

Repeat Step 3 & Continue…

➢ **Clusters are: K=2**

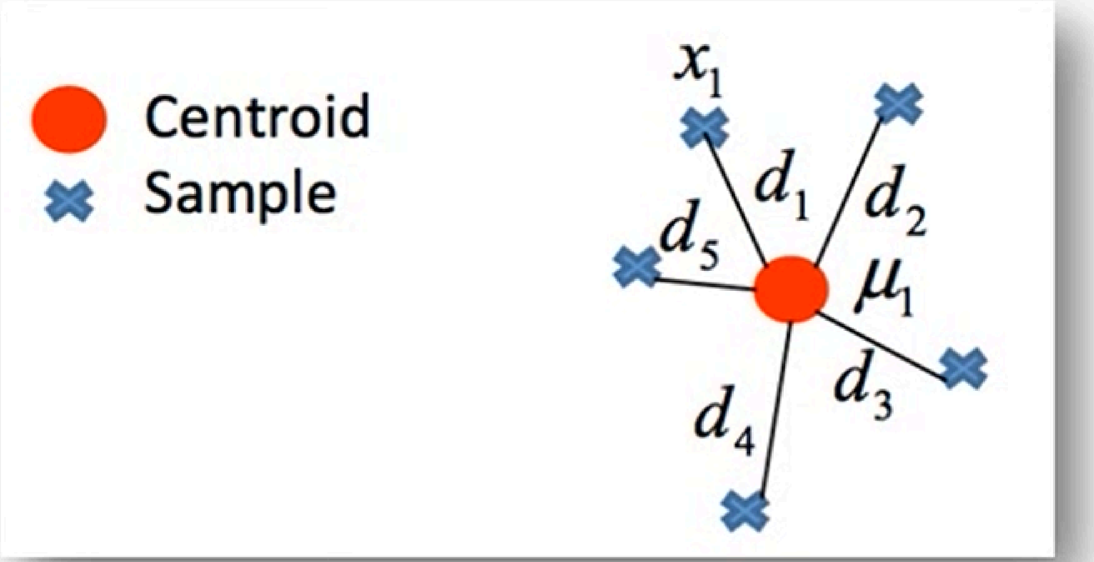
K1= {1,4,5,6,7,8,9,10,11,12}

K2 = {2,3}

Advantages & Disadvantages

Advantages of K-means

1. It is very simple to implement.
2. It is scalable to a huge data set and also faster to large datasets.
3. It adapts the new examples very frequently.
4. Generalization of clusters for different shapes and sizes.



Disadvantages of K-means

1. It is sensitive to the outliers.
2. Choosing the k values manually is a tough job.
3. As the number of dimensions increases its scalability decreases.